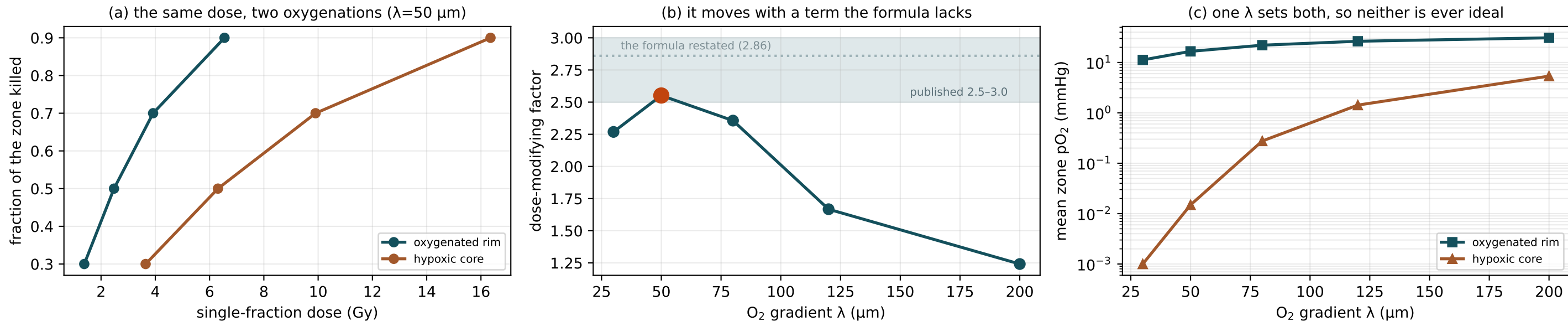


Radiation in the spatial engine: an oxygen effect measured from a population, not restated from a formula



The engine already contained the Alper-Howard-Flanders hyperbola, and `dna_channel_dose_modifying_factor` returns 2.86 from it - a number the crate documents and tests as a RESTATEMENT, so scoring it against the published band would be a guard computing its own expectation. What (b) shows is a different quantity: the dose ratio for equal kill between an oxygenated rim and a hypoxic core, read off a 60^3 population, which moves from 1.24 to 2.55 with the oxygen gradient - a parameter the formula does not contain. It reaches the published band at ONE gradient and falls below it at the engine's own zone reference. (c) is why, and why every value is a LOWER bound: one lambda sets the rim and the core together, so a gradient steep enough to make the core anoxic leaves the rim hypoxic too, and the model can never present the fully-oxic-versus-anoxic pair the published band was measured on.`